

Web Recommender Systems Project

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Week 6: Data Cleaning

Dataset: “Musical Instruments”

- ~14K ratings
- ~900 users
- ~500 items

Data **Cleaning** Steps:

- Removed missing values
- Duplicate entries
- Ensuring that all test set users are present in the training set

Dataset	Before Cleaning	After Cleaning
Train Set	11000	9913
Test Set	3000	1822

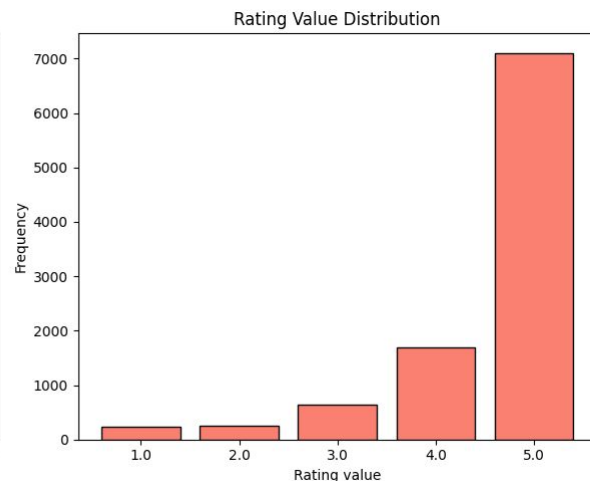
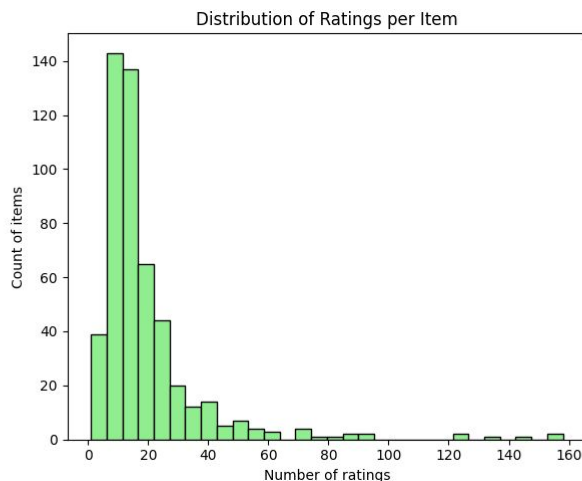
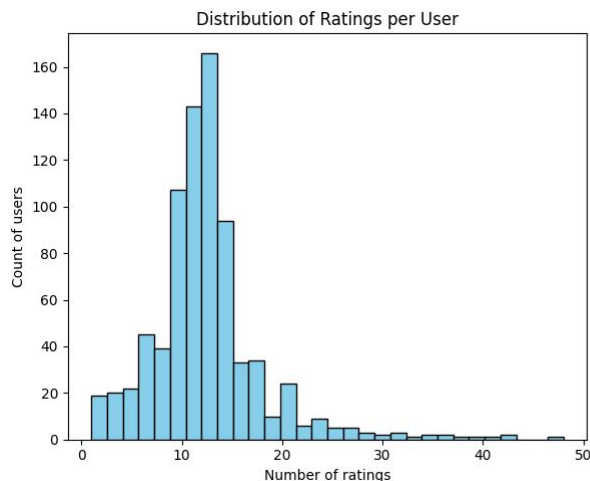
Table 1: Effect of the Data Cleaning Process on the Number of Ratings

Dataset	Before Cleaning	After Cleaning
Train Set	800	800
Test Set	437	437

Table 2: Effect of the Data Cleaning Process on the Number of Users

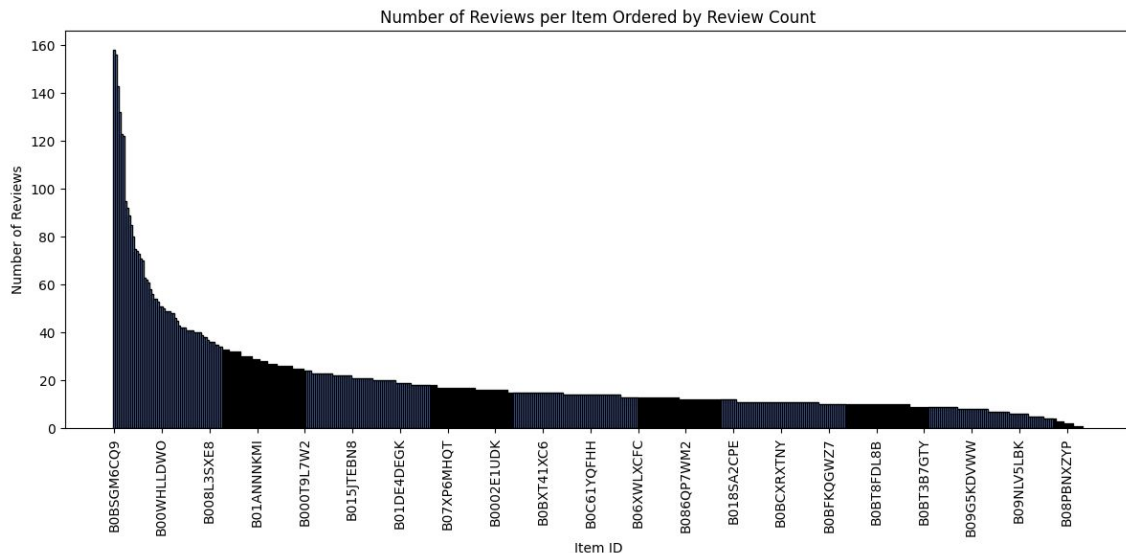
Week 6: Graphs and Analysis

The following **per-user** and **per-item** metrics were plotted from the ratings in the training set:



Week 6: Key Findings

- Distribution of ratings shows a pronounced **long-tail**
- Highly **sparse** user-item matrix: Common problem for RS



Week 7: Grid Search & Hyperparameters

Models Considered:

- **TopPop, KNNWithMeans** (user-based), **SVD**

Tuning Details:

- **User-based KNNWithMeans**
 - k: [5, 10, 20, 30, 40, 50, 70, 100]
 - Similarity measure: ['cosine', 'msd']
- **SVD**
 - n factors: [3, 5, 10, 20, 50, 100]
 - n epochs: [5, 10, 20, 50, 100]



Week 7: Grid Search Outcomes

To determine the best parameter combination, I performed a grid search with **5-fold cross-validation** on the training set.

- Metric to assess the quality: **RMSE**

Model	Best Hyperparameters	Avg RMSE
KNNWithMeans	'k': 70, 'similarity measure ': 'cosine'	0.9100
SVD	'n_factors': 5, 'n_epochs': 20	0.8440

Table 4: Best Hyperparameters and **Average** RMSE **Across the 5 Folds** for Each Model



Week 8: Test Set RMSE Evaluation

After tuning, both models were run on the full training set and used to make **predictions** on the missing ratings.

The quality of the predictions has been assessed on the **test set** using the **RMSE**:

Model	Validation RMSE	Test RMSE
KNNWithMeans	0.9100	1.0619
SVD	0.8440	0.9795

Table 5: RMSE Comparison on Test Set

Week 8: Further Metrics

RMSE can be useful, but does not consider ranking quality and does not directly reflect recommendation usefulness.

For this reason the following **metrics** were computed:

Model	Hit Rate	Precision@10	MAP@10	MRR@10	Coverage
KNNWithMeans	0.1092	0.0117	0.0084	0.0314	0.5874
SVD	0.0825	0.0083	0.0061	0.0222	0.1179
TopPop	0.2573	0.0325	0.0339	0.1187	0.0196

Table 6: Evaluation Metrics for Top-10 Recommendations **on the Test Set (ratings ≥ 4 considered relevant)**

Week 9: Description Field Preprocessing

Applied NLP preprocessing on item descriptions (after keeping only first **1000** characters):

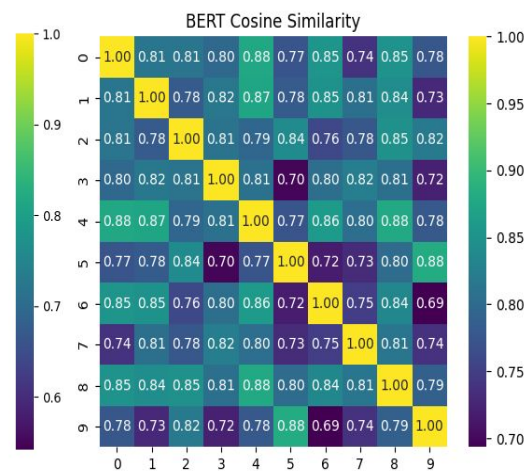
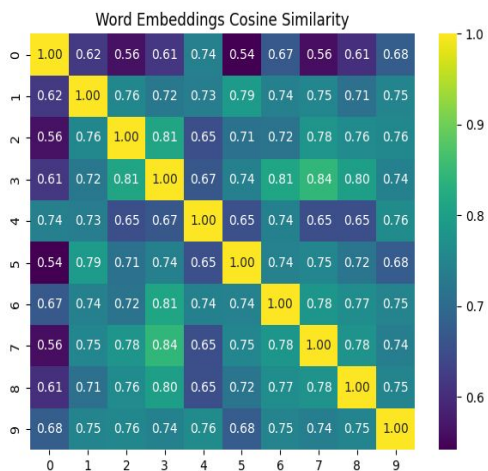
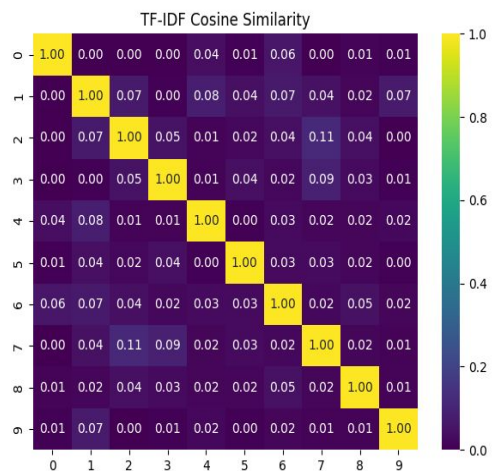
1. Tokenization and lowercasing
2. Punctuation removal
3. Stopword removal
4. Digit-containing token removal
5. Stemming (using the SnowballStemmer)

Impact: Vocabulary reduced from **51,296** to **27,243** tokens



Week 9: Vector Representations & Analysis

- **TF-IDF** (sparse, captures term overlap)
- **Word2Vec** (dense, captures semantic similarity)
- **BERT**-based (different kinds of semantic relationships)



Week 10: Building Item and User Profiles

Item Profiles:

- **Combined TF-IDF** features from title and description (total dimension: 41,550)

User Profiles:

- **Averaging** the feature vectors of the items a user rated

To make a first assessment of those profiles:

$$\text{predicted_rating} = 1 + 4 \cdot \text{similarity}$$

On 100 random ratings in the test set: High RMSE of **3.18**

Week 10: Content-Based Approach

To avoid distortions I used the cosine similarities themselves to produce ranked recommendation lists. For each user, all **unrated items were sorted by their similarity to the user's profile** vector, and the top-10 most similar items were selected as recommendations.

Model	Hit Rate	Precision@10	MAP@10	MRR@10	Coverage
Content-Based	0.0121	0.0012	0.0007	0.0022	0.0485

Table 9: Content-Based Recommender Evaluation Results

Week 11: Hybrid Strategies Overview

Hybrid approaches combine a collaborative filtering method (user-based **KNNWithMeans**) with a **content-based** method:

- **Parallel Combination:** Aggregates rankings from both models with Borda Count (giving them the same weight)
- **Switching Strategy:** Uses CF for active users (more than 5 ratings) and content-based for cold-starts
- **Pipeline Hybrid:** Candidate filtering with CB (to select the top 25 items) followed by re-ranking via CF



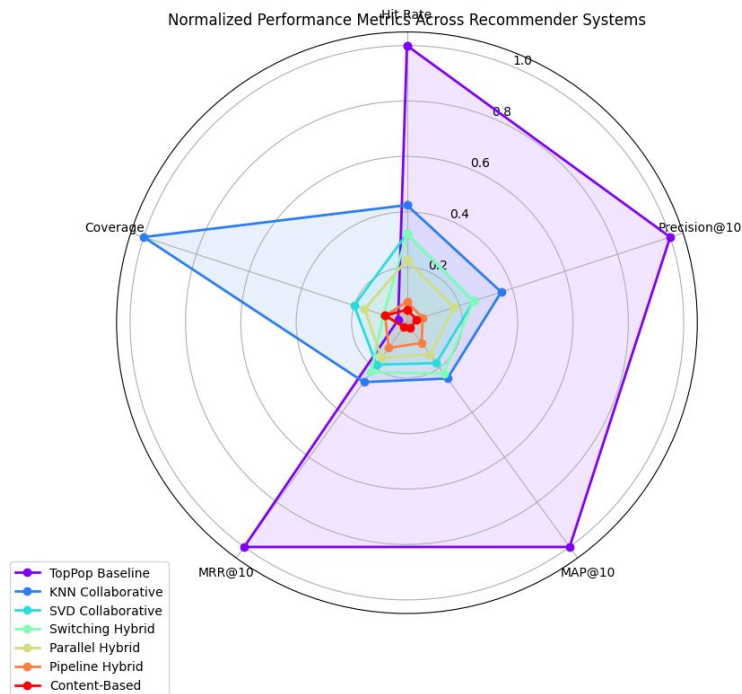
Week 11: Evaluation Outcomes

The standard **metrics** were computed on the test set:

Model	Hit Rate	Precision@10	MAP@10	MRR@10	Coverage
TopPop Baseline	0.2573	0.0325	0.0339	0.1187	0.0196
KNN Collaborative	0.1092	0.0117	0.0084	0.0314	0.5874
SVD Collaborative	0.0825	0.0083	0.0061	0.0222	0.1179
Switching Hybrid	0.0825	0.0083	0.0077	0.0262	0.0526
Parallel Hybrid	0.0583	0.0058	0.0048	0.0186	0.0944
Pipeline Hybrid	0.0194	0.0019	0.0031	0.0134	0.0486
Content-Based	0.0121	0.0012	0.0007	0.0022	0.0485

Table 10: Comparison of selected recommenders on the test set (ratings ≥ 4 considered relevant)

Week 11: Final Conclusions



TopPop performs well on many metrics but offers low coverage by recommending only popular items. **KNN**-based CF balances accuracy and diversity yet struggles with cold-start cases, while **hybrid methods** leverage content signals to boost robustness and recommend unseen items.



**Thank
you for your
attention**